Yukuan **Hu**

Ph.D. Candidate · Computational Mathematics

No. 55, Zhongguancun East Road, Haidian District, Beijing 100190, China

(+86) 18810512770 | ykhu@lsec.cc.ac.cn | https://huyukuan.github.io | GitHub | Google Scholar

Education

Academy of Mathematics and Systems Science (AMSS), Chinese Academy of Sciences (CAS) — *Beijing, China*

Ph.D. Candidate in Computational Mathematics (Major in Optimization) — *09/2019 - Present*

Supervisor: Prof. Xin Liu

School of Mathematical Sciences, Tongji University — *Shanghai, China*

B.Sc. in Mathematics and Applied Mathematics — *09/2015 - 07/2019*

Thesis advisor: Prof. Junfeng Yin

Research Interests

- **Numerical optimization**
- **Computational materials science**
- **Machine learning**

My interests focus on developing numerical optimization methods for scientific and engineering applications. Recently, I have been working on the topics in computational materials science, including strongly correlated electrons systems and structure relaxation with physical constraints, as well as machine learning-based approaches. Motivated by the ubiquitous large-scale applications, I also pay attention to the development and analysis of first-order optimization methods.

Publications & Preprints

1. The convergence properties of infeasible inexact proximal alternating linearized minimization
   Science China Mathematics, 2023, 66(10): 2385-2410 (doi, preprint)
   Yukuan Hu, Xin Liu

2. Sampling-based approaches for multimarginal optimal transport problems with Coulomb cost
   arXiv preprint, arXiv: 2306.16763 (preprint)
   Yukuan Hu, Mengyu Li, Xin Liu, Cheng Meng

3. A global optimization approach for multimarginal optimal transport problems with Coulomb cost
   SIAM Journal on Scientific Computing, 2023, 45(3): A1214–A1238 (doi, preprint)
   Yukuan Hu, Huajie Chen, Xin Liu

4. The exactness of the ℓ_1 penalty function for a class of mathematical programs with generalized complementarity constraints
   Fundamental Research, in press (doi, preprint)
   Yukuan Hu, Xin Liu

5. Force-based gradient descent method for *ab initio* atomic structure relaxation
   Physical Review B, 2022, 106(10): 104101 (10 pages) (doi, preprint)
   Yukuan Hu, Xingyu Gao, Yafan Zhao, Xin Liu, Haifeng Song

Patents & Copyrights

1. 晶体结构弛豫软件包 ProME-SuRe
   Description: a suite for crystal structure relaxation; static-link library files are available upon request
   CN Software Copyright, application in progress
   Xingyu Gao, Xin Liu, **Yukuan Hu**, Haifeng Song, Xin Chen, Yuechao Wang, Lifang Wang

2. 固定晶格体积晶体结构弛豫的计算方法及装置
   Description: computational methods and apparatus for crystal structure relaxation with a fixed unit cell volume
   CN Patent, 2023, ZL 202211210741.3
   Xingyu Gao, Xin Liu, Haifeng Song, **Yukuan Hu**, Xin Chen, Yuechao Wang, Jun Fang, Lifang Wang, Le Zhang

3. 原子结构弛豫的非单调线搜索方法及装置
   Description: computational methods and apparatus for atomic structure relaxation
   CN Patent, 2022, ZL 202111534901.5
   Xingyu Gao, Xin Liu, Haifeng Song, **Yukuan Hu**, Jun Fang, Zhen Yang, Yafan Zhao, Lifang Wang, Haifeng Liu

Presentations

10/2023	The 21st Annual Meeting of China Society for Industrial and Applied Mathematics (CSIAM 2023)	*Kunming, China*
08/2023	The 10th International Congress on Industrial and Applied Mathematics (ICIAM 2023)	*Tokyo, Japan*
08/2023	The 14th International Conference on Numerical Optimization and Numerical Linear Algebra (ICNONLA 2023)	*Taiyuan, China*
06/2023	Applied Math Ph.D. Seminar at Fudan University	*Shanghai, China*

Honors & Awards

09/2023	AMSS Special Prize of President Scholarship for Postgraduate Students (32,000 RMB)	*AMSS, CAS*
04/2023	Financial Support Program	*ICIAM 2023*
09/2022	Loo-Keng Hua Scholarship (100,000 RMB)	*AMSS, CAS*
09/2021	Loo-Keng Hua Scholarship (100,000 RMB)	*AMSS, CAS*
07/2019	Excellent Graduation Thesis	*Tongji University*
12/2018	SAS China Data Mining Champion	*SAS*

Skills

Languages	Chinese (mother tongue), English (professional fluency)
Programming	Fortran, C, MATLAB, Shell, Python, LaTeX
Parallel Computing	MPI, OpenMP

Professional Activities

Referee for Journals

- IEEE Transactions on Signal Processing